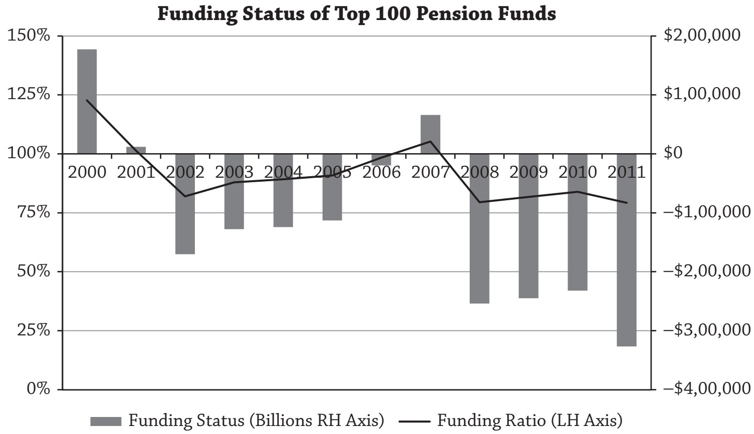

Figure 1.3

Underfunding of public pension plans is much, much worse than their corporate counterparts. Finance professors Robert Novy-Marx and Joshua Rauh estimate the underfunding problem of states and municipalities in a series of papers.[36] As of June 2009, states had accrued $5.7 trillion of retirement liabilities to their workers, yet assets in state pension plans totaled less than $2 trillion. This is a black hole of over $3 trillion, which is more than three times larger than the total outstanding publicly traded debt issued by states. Things are even worse for municipalities. Examining the largest pension plans of major cities and counties, Novy-Marx and Rauh estimate a total unfunded obligation of $7,000 per household.

Social Security, the pillar one U.S. retirement scheme, is itself underfunded. The Social Security Administration estimates that the fund will be empty in 2033, after which payroll taxes will cover only 75% of its promised obligations. The additional money required to pay all scheduled benefits is $8.6 trillion.[37] This is equivalent to 80% of total U.S. Treasury debt ($11 trillion at year-end 2012). Pension promises can be extremely expensive, especially when payouts are linked to inflation, as they are with Social Security. MetLife estimates that an annuity

[36] See Novy-Marx (2009, 2011a, 2011b). In certain circumstances, Bohn (2011) shows that underfunding a public pension plan can be optimal—but he assumes that there will always be municipal revenue available (at some time in the future) to meet the pension obligations.

[37] Numbers from the 2012 Annual Report of the Board of Trustees of the Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds. My colleague Stephen Zeldes at Columbia Business School argues that the shortfall is less if market values of Social Security liabilities would be used (see Geanakoplos and Zeldes (2011)), but there is still a shortfall. The calculation depends crucially on the correlation of labor income with equity returns, a topic that we cover in chapter 5. Note also that Social Security liabilities do not have the "full faith and credit" backing of the U.S. government and are officially not a U.S. government liability.